

Brazilian E-Commerce Business insights

Data Analysis Project

Purpose:

Try to understand what drives sales, which products are most profitable, and who the most valuable customers are, and how can the business increase sales based on data collected.

Scope:

The following project focuses on mock-data collected from Kaggle, trying to derive key insights from collected data. First I will focus on properly sorting and filtering the data to access important information using Bigquery SQL platform. Next I will use Python and pandas to calculate crucial statistical metrics, alongside modeling spending trends and understanding what makes an ideal customer. Lastly, learned patterns and information will be transferred onto a visualisation tool to mimic the 'Share' stage of data analysis.

Activity	Description
Sort and filter collected data	Use Kaggle to access the data and then BigQuery to use structured queries to extract essential information that will later be processed using pandas.
calculate statistical metrics, understand what makes an ideal customer (key characteristics)	Use Python and pandas to derive metric-based insights into how customers behave and what items are most frequently bought and identify outliers. Organise data to use for an easy import further into the visualisation phase
present data in an appealing and easy to understand way	Create a slide deck/presentation to visualise and highlight main themes and observations derived from analysed data. suggest solutions or potential roadblocks

This project does not include:

The analysis does not concern itself with comprehending customer behaviour or with a complex profiling of its user base (sex, age, earnings) and how they might affect their spending patterns.

Deliverables:

Deliverable	Description/ Details
explanation of what drives sales	aforementioned statistical analysis
which products are most profitable	find the top 3 products either most often sold or generating the biggest sales
who are the most valuable customers	analyse simple customer trends, (what characteristics make a customer spend more: what payment method, location, etc)
Provide business oriented suggestions	provide 3 ideas on how sales can be increased

Schedule Overview / Major Milestones:

Milestone	Expected Completion Date	Description/Details
<i>Finish Activity 1</i>	<i>04.02.2026</i>	<i>Derive all necessary data from the dataset</i>
<i>Finish Activity 2</i>	<i>11.04.2026</i>	<i>compute all needed metrics</i>
<i>Finish Activity 3</i>	<i>18.04.2026</i>	<i>adjust the presentation keeping in mind all described deliverables</i>
<i>Fully ship analysis project</i>	<i>25.04.2026</i>	<i>If needed revisit uncompleted deliverables and adjust the visualization</i>

Estimated date for completion: 25.04.2026

Provided there are no major setbacks or roadblocks.